Calculus I
Homework
Areas and integrals

1. Evaluate the difference quotient and simplify your answer.

   (a) $\dfrac{f(2+h)-f(2)}{h}$, where $f(x)=x^2-x-1$.

   (b) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x)=x^2$.

   (c) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x)=x^3$.

   (d) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x)=x^4$.

   (e) $\dfrac{f(x)-f(a)}{x-a}$, where $f(x)=\frac{1}{x}$.

   (f) $\dfrac{f(x)-f(1)}{x-1}$, where $f(x)=\frac{x-1}{x+1}$.

2. Write down a formula for a function whose graphs is given below. The graphs are up to scale. Please note that there is more than one way to write down a correct answer.

(b)

(c)

(d)

3. Write down formulas for function whose graphs are as follows. The graphs are up to scale. All arcs are parts of circles.

   (a)

4. Evaluate the difference quotient and simplify your answer.

   (a) $\dfrac{f(2+h)-f(2)}{h}$, where $f(x)=x^2-x-1$.

   (b) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x)=x^2$.

   (c) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x)=x^3$.

   (d) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x)=x^4$.

   (e) $\dfrac{f(x)-f(a)}{x-a}$, where $f(x)=\frac{1}{x}$.

   (f) $\dfrac{f(x)-f(1)}{x-1}$, where $f(x)=\frac{x-1}{x+1}$.

5. Find the implied domain of the function.

   (a) $f(x)=\dfrac{x+4}{x^2-4}$.

   (b) $f(x)=\dfrac{2x^3-5}{x^2+5x+6}$.

   (c) $f(t)=\sqrt[3]{3t-1}$.

   (d) $g(t)=\sqrt{5-t}-\sqrt{1+t}$.

   (e) $h(x)=\dfrac{1}{\sqrt[6]{x^2-7x}}$.

   (f) $f(u)=\dfrac{u+1}{1+\frac{1}{u+1}}$.

   (g) $F(x)=\sqrt{10-\sqrt{x}}$.

6. Find the implied domain of the function.

   (a) $f(x)=\dfrac{x+4}{x^2-4}$.

   (b) $f(x)=\dfrac{2x^3-5}{x^2+5x+6}$.

   (c) $f(t)=\sqrt[3]{3t-1}$.

   (d) $g(t)=\sqrt{5-t}-\sqrt{1+t}$.

   (e) $h(x)=\dfrac{1}{\sqrt[6]{x^2-7x}}$.

   (f) $f(u)=\dfrac{u+1}{1+\frac{1}{u+1}}$.

   (g) $F(x)=\sqrt{10-\sqrt{x}}$.

7. Compute the composite functions $(f\circ g)(x)$, $(g\circ f)(x)$. Simplify your answer to a single fraction. Find the domain of the composite function.

   (a) $f(x)=\dfrac{x+2}{x-2}, g(x)=\dfrac{x-1}{x+2}$.

   (b) $f(x)=\dfrac{x+1}{3x-2}, g(x)=\dfrac{x-2}{x-1}$.

   (c) $f(x)=\dfrac{2x+1}{3x-1}, g(x)=\dfrac{x-2}{2x-1}$.

   (d) $f(x)=\dfrac{x+1}{x-2}, g(x)=\dfrac{x+2}{2x-1}$.

   (e) $f(x)=\dfrac{5x+1}{4x-1}, g(x)=\dfrac{4x-1}{3x+1}$.

   (f) $f(x)=\dfrac{3x-5}{x-2}, g(x)=\dfrac{x-2}{x-4}$.

   (g) $f(x)=\dfrac{x-3}{x+2}, g(y)=\dfrac{y+3}{y-4}$.

8. Find the functions $f\circ g$, $g\circ f$, $f\circ f$ and $g\circ g$ and their implied domains. The answer key has not been proofread, use with caution.

   (a) $f(x)=x^2+1, g(x)=x+1$.

   (b) $f(x)=\sqrt{x+1}, g(x)=x+1$.

   (c) $f(x)=2x, g(x)=\tan x$.
   In this subproblem, you are not required to find the domain.

   (d) $f(x)=\dfrac{x+1}{x-1}, g(x)=\dfrac{x-1}{x+1}$.

9. Convert from degrees to radians.

(a) $15°$.
(b) $30°$.
(c) $36°$.
(d) $45°$.
(e) $60°$.
(f) $75°$.
(g) $90°$.
(h) $120°$.
(i) $135°$.
(j) $150°$.
(k) $180°$.
(l) $225°$.
(m) $270°$.
(n) $305°$.
(o) $360°$.
(p) $405°$.
(q) $1200°$.
(r) $-900°$.
(s) $-2014°$.

10. Convert from radians to degrees. The answer key has not been proofread, use with caution.

(a) 4π.
(b) $-\frac{7}{6}\pi$.
(c) $\frac{7}{12}\pi$.
(d) $\frac{4}{3}\pi$.
(e) $-\frac{3}{8}\pi$.
(f) 2014π.
(g) 5.
(h) -2014.

11. Prove the trigonometry identities.

(a) $\sin\theta\cot\theta = \cos\theta$.
(b) $(\sin\theta + \cos\theta)^2 = 1 + \sin(2\theta)$.
(c) $\sec\theta - \cos\theta = \tan\theta\sin\theta$.
(d) $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$.
(e) $\cot^2\theta + \sec^2\theta = \tan^2\theta + \csc^2\theta$.
(f) $2\csc(2\theta) = \sec\theta\csc\theta$.
(g) $\tan(2\theta) = \dfrac{2\tan\theta}{1-\tan^2\theta}$.
(h) $\dfrac{1}{1-\sin\theta} + \dfrac{1}{1+\sin\theta} = 2\sec^2\theta$.
(i) $\tan\alpha + \tan\beta = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$.
(j) $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1-\tan\alpha\tan\beta}$.
(k) $\sin(3\theta) + \sin\theta = 2\sin(2\theta)\cos\theta$.
(l) $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$.
(m) $1 + \tan^2\theta = \sec^2\theta$.
(n) $1 + \csc^2\theta = \cot^2\theta$.
(o) $2\cos^2(2x) = 2\sin^4\theta + 2\cos^4\theta - \sin^2(2\theta)$.
(p) $\dfrac{1+\tan\left(\frac{\theta}{2}\right)}{1-\tan\left(\frac{\theta}{2}\right)} = \tan\theta + \sec\theta$.

12. Find all values of x in the interval $[0, 2\pi]$ that satisfy the equation.

(a) $2\cos x - 1 = 0$.
(b) $\sin(2x) = \cos x$.
(c) $\sqrt{3}\sin x = \sin(2x)$. .
(d) $2\sin^2 x = 1$.
(e) $2 + \cos(2x) = 3\cos x$.
(f) $2\cos x + \sin(2x) = 0$.
(g) $2\cos^2 x - (1+\sqrt{2})\cos x + \dfrac{\sqrt{2}}{2} = 0$.
.
(h) $|\tan x| = 1$.
(i) $3\cot^2 x = 1$.
(j) $\sin x = \tan x$.

13. Evaluate the limits. Justify your computations.

(a) $\lim\limits_{x\to 2} 2x^2 - 3x - 6$.
(b) $\lim\limits_{x\to -1} \dfrac{x^4 - x}{x^2 + 2x + 3}$.
(c) $\lim\limits_{x\to -1} \dfrac{1}{x^2 - 3x + 2}$.
(d) $\lim\limits_{x\to -2} \sqrt{x^4 + 16}$.
(e) $\lim\limits_{x\to 8} (1 + \sqrt[3]{x})(2 - x)$.

14. Evaluate the limit if it exists.

(a) $\lim\limits_{x\to 2} \dfrac{x^2 - 5x + 6}{x - 2}$.
(b) $\lim\limits_{x\to 3} \dfrac{x^2 - 3x}{x^2 - 2x - 3}$.
(c) $\lim\limits_{x\to -2} \dfrac{2x^2 + x - 6}{x^2 - 4}$

(d) $\lim_{x\to 2} \dfrac{x^2-5x-6}{x-2}$.

(e) $\lim_{x\to -1} \dfrac{x^2-3x}{x^2-2x-3}$.

(f) $\lim_{x\to -2} \dfrac{x^2-4}{2x^2+5x+2}$.

(g) $\lim_{x\to -1} \dfrac{2x^2+3x+1}{3x^2-2x-5}$.

(h) $\lim_{x\to -4} \dfrac{x^2+7x+12}{x^2+6x+8}$.

(i) $\lim_{h\to 0} \dfrac{(-3+h)^2-9}{h}$.

(j) $\lim_{h\to 0} \dfrac{(-2+h)^3+8}{h}$.

(k) $\lim_{x\to -3} \dfrac{x+3}{x^3+27}$.

(l) $\lim_{x\to 1} \dfrac{x^4-1}{x^3-1}$.

(m) $\lim_{h\to 0} \dfrac{\sqrt{4+h}-2}{h}$.

(n) $\lim_{x\to 3} \dfrac{\sqrt{5x+1}-4}{x-3}$.

(o) $\lim_{x\to -3} \dfrac{\sqrt{x^2+16}-5}{x+3}$.

(p) $\lim_{x\to -3} \dfrac{\frac{1}{3}+\frac{1}{x}}{3+x}$.

(q) $\lim_{x\to -2} \dfrac{x^2+4x+4}{x^4-16}$.

(r) $\lim_{x\to 0} \dfrac{\sqrt{1+x}-\sqrt{1-x}}{x}$.

(s) $\lim_{x\to 0} \left(\dfrac{1}{x}-\dfrac{1}{x^2+x}\right)$.

(t) $\lim_{x\to 9} \dfrac{3-\sqrt{x}}{9x-x^2}$.

(u) $\lim_{h\to 0} \dfrac{(2+h)^{-1}-2^{-1}}{h}$.

(v) $\lim_{x\to 0} \left(\dfrac{1}{x\sqrt{1+x}}-\dfrac{1}{x}\right)$.

(w) $\lim_{h\to 0} \dfrac{(x+h)^3-x^3}{h}$.

(x) $\lim_{h\to 0} \dfrac{\frac{1}{(x+h)^2}-\frac{1}{x^2}}{h}$.

(y) $\lim_{h\to 0} \dfrac{\frac{1}{(2+h)^2}-\frac{1}{4}}{h}$.

(z) $\lim_{h\to 0} \dfrac{\frac{1}{(1+h)^2}-1}{h}$.

15. Find the (implied) domain of $f(x)$. Extend the definition of f at $x=3$ to make f continuous at 3.

(a) $f(x)=\dfrac{x^2-x-6}{x-3}$.

(b) $f(x)=\dfrac{x^3-27}{x^2-9}$.

16. Use the Intermediate Value Theorem to show that there is a real number solution of the given equation in the specified interval.

(a) $x^5+x-3=0$ where $x\in(1,2)$.

(b) $\sqrt[4]{x}=1-x$ where $x\in\mathbb{R}$ (i.e., x is an arbitrary real number).

(c) $\cos x=2x$, where $x\in(0,1)$.

(d) $\sin x=x^2-x-1$, where $x\in\mathbb{R}$ (i.e., x is an arbitrary real number).

(e) $\cos x=x^4$, where $x\in\mathbb{R}$ (i.e., x is an arbitrary real number).

(f) $x^5-x^2+x+3=0$, where $x\in\mathbb{R}$.

17.

(a) i. Solve the equation $x^2+13x+41=1$.

ii. Use the intermediate value theorem to prove that the equation $x^2+13x+41=\sin x$ has at least two solutions, lying between the two solutions to 17.a.i.

(b) i. Solve the equation $x^2-15x+55=1$.

ii. Use the intermediate value theorem to prove that the equation $x^2-15x+55=\cos x$ has at least two solutions, lying between the two solutions to the equation in the preceding item.

18. For which values of x is f continuous?

- $f(x)=\begin{cases} 0 & \text{if } x \text{ is rational} \\ 1 & \text{if } x \text{ is irrational} \end{cases}$
- $f(x)=\begin{cases} 0 & \text{if } x \text{ is rational} \\ x & \text{if } x \text{ is irrational} \end{cases}$

19. Show that $f(x)$ is continuous at all irrational points and discontinuous at all rational ones.

$$f(x)=\begin{cases} \frac{1}{q^2} & \text{if } x \text{ is rational and } x=\frac{p}{q} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

where in the first item p, q are relatively prime integers (i.e., integers without a common divisor).

20. Show the following limits do not exist and compute whether they evaluate to ∞, $-\infty$, or neither.

(a) $\lim_{x\to 3^+} \frac{x^2+x-1}{x^2-2x-3}$.

(b) $\lim_{x\to 3^-} \frac{x^2+x-1}{x^2-2x-3}$.

(c) $\lim_{x\to 1^+} \frac{x^2+1}{\sqrt{x^2+3}-2}$.

(d) $\lim_{x\to 1^-} \frac{x^2+1}{\sqrt{x^2+3}-2}$.

(e) $\lim_{x\to 2^+} \frac{\sqrt{x^3-8}}{-x^2+x+2}$.

(f) $\lim_{x\to -1^+} \frac{\sqrt[3]{x^2+2x+1}}{x^2-2x-3}$.

21. Find the limit or show that it does not exist. If the limit does not exist, indicate whether it is $\pm\infty$, or neither. The answer key has not been proofread, use with caution.

(a) $\lim_{x\to\infty} \frac{x-2}{2x+1}$.

(b) $\lim_{x\to\infty} \frac{1-x^2}{x^3-x-1}$.

(c) $\lim_{x\to-\infty} \frac{x-2}{x^2+5}$.

(d) $\lim_{x\to-\infty} \frac{3x^3+2}{2x^3-4x+5}$.

(e) $\lim_{x\to\infty} \frac{\sqrt{x}+x^2}{\sqrt{x}-x^2}$.

(f) $\lim_{x\to\infty} \frac{3-x\sqrt{x}}{2x^{\frac{3}{2}}-2}$.

(g) $\lim_{x\to\infty} \frac{(2x^2+3)^2}{(x-1)^2(x^2+1)}$.

(h) $\lim_{x\to\infty} \frac{x^2-3}{\sqrt{x^4+3}}$.

(i) $\lim_{x\to-\infty} \frac{\sqrt{x^2+1}}{x+1}$.

(j) $\lim_{x\to\infty} \frac{\sqrt{16x^6-3x}}{x^3+2}$.

(k) $\lim_{x\to-\infty} \frac{\sqrt{16x^6-3x}}{x^3+2}$.

(l) $\lim_{x\to\infty} \frac{\sqrt{3x^2+2x+1}}{x+1}$.

(m) $\lim_{x\to\infty} \sqrt{4x^2+x}-2x$.

(n) $\lim_{x\to-\infty} x+\sqrt{x^2+3x}$.

(o) $\lim_{x\to\infty} \sqrt{x^2+2x}-\sqrt{x^2-2x}$.

(p) $\lim_{x\to-\infty} \sqrt{x^2+x}-\sqrt{x^2-x}$.

(q) $\lim_{x\to\infty} \sqrt{x^2+ax}-\sqrt{x^2+bx}$.

(r) $\lim_{x\to\infty} \cos x$.

(s) $\lim_{x\to\infty} \frac{x^4+x}{x^3-x+2}$.

(t) $\lim_{x\to\infty} \sqrt{x^2+1}$.

(u) $\lim_{x\to-\infty} (x^4+x^5)$.

(v) $\lim_{x\to-\infty} \frac{\sqrt{1+x^6}}{1+x^2}$.

(w) $\lim_{x\to\infty} (x-\sqrt{x})$.

(x) $\lim_{x\to\infty} (x^2-x^3)$.

(y) $\lim_{x\to\infty} x\sin x$.

(z) $\lim_{x\to\infty} \sqrt{x}\sin x$.

22. Find the horizontal and vertical asymptotes of the graph of the function. If a graphing device is available, check your work by plotting the function.

(a) $y=\frac{2x}{\sqrt{x^2+x+3}-3}$.

(b) $y=\frac{3x^2}{\sqrt{x^2+2x+10}-5}$.

(c) $y=\frac{3x+1}{x-2}$.

(d) $y=\frac{x^2-1}{2x^2-3x-2}$.

(e) $y=\frac{2x^2-2x-1}{x^2+x-2}$.

(f) $f(x)=\frac{-5x^2-3x+5}{x^2-2x-3}$

(g) $y=\frac{1+x^4}{x^2-x^4}$.

(h) $y=\frac{x^3-x}{x^2-7x+6}$.

(i) $y=\frac{x-9}{\sqrt{4x^2+3x+3}}$.

(j) $y=\frac{\sqrt{x^2+1}-x}{x}$.

(k) $f(x)=\frac{x}{\sqrt{x^2+3}-2x}$

23. Find the inverse function. You are asked to do the algebra only; you are not asked to determine the domain or range of the function or its inverse.

(a) $f(x)=3x^2+4x-7$, where $x\geq -\frac{2}{3}$.

(b) $f(x)=2x^2+3x-5$, where $x\geq -\frac{3}{4}$.

(c) $f(x)=\frac{2x+5}{x-4}$, where $x\neq 4$.

(d) $f(x)=\frac{3x+5}{2x-4}$, where $x\neq 2$.

(e) $f(x)=\frac{5x+6}{4x+5}$, where $x\neq -\frac{5}{4}$.

(f) $f(x)=\frac{2x-3}{-3x+4}$, where $x\neq \frac{4}{3}$..

24. Find the inverse function and its domain.

(a) $y = \ln(x+3)$.
(b) $y = 4\ln\,(x-3) - 4$.
(c) $y = 2\ln\,(-2x+4) + 1$
(d) $f(x) = e^{x^3}$.
(e) $y = (\ln x)^2$, $x \geq 1$.
(f) $y = \dfrac{e^x}{1+2e^x}$.
(g) $f(x) = 2^{2x} + 2^x - 2$.

25. Find the exact value of each expression.

(a) $\log_5 125$.
(b) $\log_3 \dfrac{1}{27}$.
(c) $\ln\left(\dfrac{1}{e}\right)$.
(d) $\log_{10} \sqrt{10}$.
(e) $e^{\ln 4.5}$.
(f) $\log_{10} 0.0001$.
(g) $\log_{1.5} 2.25$.
(h) $\log_5 4 - \log_5 500$.
(i) $\log_2 6 - \log_2 15 + \log_2 20$.
(j) $\log_3 100 - \log_3 18 - \log_3 50$.
(k) $e^{-2\ln 5}$.
(l) $\ln\left(\ln e^{e^{10}}\right)$.

26. Use the definition of a logarithm to evaluate each of the following without using a calculator.

(a) $\log_2 16$
(b) $\log_3 \left(\dfrac{1}{9}\right)$
(c) $\log_{10} 1000$
(d) $\log_6 36^{-\frac{2}{3}}$
(e) $\log_2(8\sqrt{2})$
(f) $\log_7 \left(\dfrac{49^x}{343^y}\right)$

27. Express each of the following as a single logarithm.

(a) $\ln 4 + \ln 6 - \ln 5$
(b) $2\ln 2 - 3\ln 3 + 4\ln 4$
(c) $\ln 36 - 2\ln 3 - 3\ln 2$

28. Solve each equation for x. If available, use a calculator to give an ($\approx$) answer in decimal notation. If available, use a calculator to verify your approximate solutions.

(a) $e^{7-4x} = 7$.
(b) $\ln(2x-9) = 2$.
(c) $\ln(x^2-2) = 3$.
(d) $2^{x-3} = 5$.
(e) $\ln x + \ln(x-1) = 1$.
(f) $e^{2x+1} = t$.
(g) $\log_2(mx) = c$.
(h) $e - e^{-2x} = 1$.
(i) $8(1+e^{-x})^{-1} = 3$.
(j) $\ln(\ln x) = 1$.
(k) $e^{e^x} = 10$.
(l) $\ln(2x+1) = 3 - \ln x$.
(m) $e^{2x} - 4e^x + 3 = 0$.
(n) $e^{4x} + 3e^{2x} - 4 = 0$.
(o) $e^{2x} - e^x - 6 = 0$.
(p) $4^{3x} - 2^{3x+2} - 5 = 0$.
(q) $3 \cdot 2^x + 2\left(\frac{1}{2}\right)^{x-1} - 7 = 0$.

29. Compute the derivative.

(a) $f(x) = 2^{2015}$.
(b) $f(x) = \pi^{2015}$.
(c) $f(x) = 2 - \dfrac{2}{3}x$.
(d) $f(x) = \dfrac{3}{4}x^8$.
(e) $f(x) = x^3 - 4x + 6$.
(f) $f(t) = \dfrac{1}{2}t^6 - 3t^4 + t$.

(g) $g(x) = x^2(1-2x)$.

(h) $h(x) = (x-2)(2x+3)$.

(i) $f(x) = 2x^{-\frac{3}{4}}$.

(j) $f(x) = cx^{-6}$.

(k) $A(x) = -\dfrac{12}{x^5}$.

30. (a) Given that $f(0) = 5$, $f'(0) = -1$, $g(0) = -4$, $g'(0) = 1$ and $h(x) = f(x)g(x)$, find the derivative $h'(0)$.

    (b) Given that $f(2) = -3$, $f'(2) = 2$, $g(2) = 5$, $g'(2) = 1$ and $h(x) = f(x)g(x)$, find the derivative $h'(2)$.

    (c) Given that $f(0) = 5$, $f'(0) = -1$, $g(0) = -4$, $g'(0) = 1$ and $h(x) = \frac{f(x)}{g(x)}$, find the derivative $h'(0)$.

    (d) Given that $f(1) = 2$, $f'(1) = -1$, $g(1) = -3$, $g'(1) = 1$, $h(1) = 0$, $h'(1) = 1$ and $j(x) = f(x)g(x)h(x)$, find the derivative $j'(1)$.

31. Compute the derivative.

(a) $y = x^{\frac{5}{3}} - x^{\frac{2}{3}}$.

(b) $f(x) = \sqrt{x} - x$.

(c) $y = \sqrt{x}(x-1)$.

(d) $f(x) = (2x+1)^2$.

(e) $f(x) = 4\pi x^2$.

(f) $y = \dfrac{x^2+4x+3}{\sqrt{x}}$.

(g) $y = \dfrac{\sqrt{x}+x}{x^2}$.

(h) $f(x) = \left(x + x^{-1}\right)^3$.

(i) $f(x) = \sqrt{2x} + \sqrt{5x}$.

(j) $y = \sqrt[5]{x} + 4\sqrt{x^5}$.

(k) $y = \left(\sqrt{x} + \dfrac{1}{\sqrt[3]{x}}\right)^2$.

(l) $f(x) = (1+2x^2)(x-x^2)$.

(m) $f(x) = \dfrac{x^4 - 5x^3 + \sqrt{x}}{x^2}$.

(n) $f(x) = (2x^3+3)(x^4-2x)$.

(o) $f(x) = (1+x+x^2)(2-x^4)$.

(p) $g(y) = \left(\dfrac{1}{y^2} - \dfrac{3}{y^4}\right)(y+5y^3)$.

(q) $f(x) = (x^3-2x)(x^{-4}+x^{-2})$.

(r) $f(x) = \dfrac{1+2x}{3-4x}$.

32. Compute the derivative (with respect to the implied variable).

(a) $f(x) = \dfrac{x-3}{x+3}$.

(b) $y = \dfrac{x^3}{1-x^2}$.

(c) $y = \dfrac{x+1}{x^3+x-2}$.

(d) $y = \dfrac{x-1}{x^3+x-2}$.

(e) $f(x) = \dfrac{x+1}{x^3+1}$.

(f) $y = \dfrac{x^3 - 2x\sqrt{x}}{x}$.

(g) $y = \dfrac{t}{(t-1)^2}$.

(h) $y = \dfrac{t^2+2}{t^4-3t^2+1}$.

(i) $g(t) = \dfrac{t-\sqrt{t}}{t^{\frac{1}{3}}}$.

(j) $y = ax^2 + bx + c$.

(k) $y = A + \dfrac{B}{x} + \dfrac{C}{x^2}$.

(l) $f(t) = \dfrac{2t}{2+\sqrt{t}}$.

(m) $y = \dfrac{cx}{1+cx}$.

(n) $y = \sqrt[3]{t}(t^2+t+t^{-1})$.

(o) $y = \dfrac{u^6 - 2u^3 + 5}{u^2}$.

(p) $f(x) = \dfrac{ax+b}{cx+d}$.

(q) $f(x) = \dfrac{1+x}{1+\frac{2}{x}}$.

(r) $f(x) = \dfrac{1+x}{1+\frac{3}{x}}$.

(s) $f(x) = \dfrac{x}{x+\frac{c}{x}}$.

33. Compute the derivative.

(a) $f(x) = 2x^3 - 3\cos x$.

(b) $f(x) = \sqrt{x}\cos x$.

(c) $f(x) = \sin x + \dfrac{1}{3}\cot x$.

(d) $y = 2\sec x - \csc x$.

(e) $y = \dfrac{1+\sin^2\theta}{\cos^3\theta}$.

(f) $g(t) = 4\sec t + \tan t - \csc t + 3\cot t$.

(g) $y = c\cos t + t^2 \sin t$.

(h) $y = u(a\cos u + b\cot u)$.

(i) $y = \dfrac{x}{2-\tan x}$.

(j) $y = \sin\theta\cos\theta$.

(k) $f(\theta) = \dfrac{\sec\theta}{1+\sec\theta}$.

(l) $y = \dfrac{\cos x}{1-\sin x}$.

(m) $y = \dfrac{t\sin t}{1+t}$.

(n) $y = \dfrac{1-\sec x}{\tan x}$.

(o) $h(\theta) = \theta\csc\theta - \cot\theta$.

(p) $y = x^2 \sin x \tan x$.

34. Differentiate.

(a) $\tan x$.

(b) $\cot x$.

(c) $\sec x$.

(d) $\csc x$.

(e) $\sec x \tan x$.

(f) $\sec x + \tan x$.

(g) $\sec^2 x$.

(h) $\csc^2 x$.

(i) $f(x) = (\sec x)e^x$.

(j) $f(x) = (\tan x)e^x$.

(k) $\dfrac{\sin x}{x}$.

(l) $\dfrac{\sin x}{e^x}$.

(m) $x(\cos x)e^x$.

(n) $\dfrac{e^x}{\tan x}$.

(o) $\dfrac{e^x}{\sec x} + \sec x$.

35. Compute the derivative using the chain rule.

(a) $f(x) = \sqrt{1+x^2}$

(b) $f(x) = \sqrt{3x^2 - x + 2}$.

(c) $f(x) = \dfrac{x}{\sqrt{1+\frac{2}{x^2}}}$.

(d) $f(x) = \sqrt{1-\sqrt{x}}$.

(e) $y = (\cos x)^{\frac{1}{2}}$

(f) $f(x) = \sin^3 x$.

(g) $y = (1+\cos x)^2$.

(h) $f(x) = \dfrac{1}{\sin^3 x}$.

(i) $f(x) = \sqrt[3]{4+3\tan x}$.

(j) $f(x) = (\cos x + 3\sin x)^4$.

(k) $y = \sin\left(\sqrt{x}\right)$

(l) $y = \cos(4x)$

(m) $\sec^2(3x^2)$.

(n) $\csc^2(3x^2)$.

(o) e^{2x}.

(p) e^{-x^2}

(q) $e^{\sqrt{x}}$

(r) $f(x) = e^{-\frac{1}{x}}$.

(s) 5^x.

(t) e^{2^x}.

(u) 2^{3^x}.

(v) 3^{2^x}.

(w) $y = \sqrt{\sec(4x)}$

(x) $y = x^2\tan(5x)$

(y) $y = \dfrac{1+\sin\left(x^2\right)}{1+\cos\left(x^2\right)}$.

36. Compute the derivative.

(a) $f(x) = (x^4 + 3x^2 - 2)^5$.

(b) $f(x) = (4x - x^2)^{100}$.

(c) $f(x) = (2x-3)^4(x^2+x+1)^5$.

(d) $f(x) = (x^2+1)^3(x^2+2)^6$.

(e) $f(x) = (3x-1)^4(2x+1)^{-3}$.

(f) $f(x) = \dfrac{1}{1+x^2}$.

(g) $f(x) = \left(\dfrac{x^2+1}{x^2-1}\right)^3$.

(h) $f(x) = (x+1)^{\frac{2}{3}}(2x^2-1)^3$.

(i) $f(x) = \sqrt{1-2x}$.

(j) $f(x) = \sqrt{\dfrac{x^2+1}{x^2+4}}$.

(k) $f(x) = 3\cot(2x)$.

(l) $f(x) = \dfrac{1}{(1+\sec x)^2}$.

(m) $f(x) = \sqrt[3]{1+\tan x}$.

(n) $f(x) = \cos(2+x^3)$.

(o) $f(x) = \cos\left(\dfrac{1}{x}\right)\sin(x^2)$.

(p) $f(x) = x\sec(kx)$.

37. Differentiate.

(a) $f(x) = \sin(\tan(2x))$.
(b) $f(x) = \sec^2(mx)$.
(c) $f(x) = \sec^2 x + \tan^2 x$.
(d) $f(x) = x \sin\left(\frac{1}{x}\right)$.
(e) $f(x) = \left(\frac{1-\cos(2x)}{1+\cos(2x)}\right)^4$.
(f) $f(x) = \sqrt{\frac{x}{x^2+4}}$.
(g) $f(t) = \cot^2(\sin t)$.
(h) $f(x) = \left(ax + \sqrt{x^2+b^2}\right)^{-2}$.
(i) $f(x) = \left(x^2 + (1-3x)^5\right)^3$.
(j) $f(x) = \sin(\sin(\sin x))$.
(k) $f(x) = \sqrt{x + \sqrt{x}}$.
(l) $f(x) = \sqrt{x + \sqrt{x + \sqrt{x}}}$.
(m) $f(x) = (2r\sin(rx) + n)^p$.
(n) $f(x) = \cos^4(\sin^3 x)$.
(o) $f(x) = \cos\sqrt{\sin(\tan(\pi x))}$.
(p) $f(x) = \left(x + (x + \sin^2 x)^3\right)^4$.

38. Compute the second derivative.

(a) $f(x) = \sin(-5x)$.
(b) $f(x) = \cot(2x)$.
(c) $f(x) = e^{-3x}$.
(d) $f(x) = e^{\frac{1}{x}}$.
(e) $f(x) = e^{\sqrt{x}}$.
(f) $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
(g) $f(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

39. Compute the derivative.

(a) $\ln(4x)$
(b) $\ln(-13x)$
(c) $\log_2(5x)$
(d) $\log_{10}(-3x)$
(e) $x^6\ln(2x)$
(f) $x^4\ln(2x)$
(g) $\ln\left(x^4\right)$
(h) $(\ln(x))^4$
(i) $\ln(7x+1)$
(j) $\ln(-6x+2)$
(k) $\ln\left(\frac{3x-2}{-2x+3}\right)$
(l) $\ln\left(\frac{5x-4}{-x-5}\right)$
(m) $\ln\left(\frac{3x+1}{4x-5}\right)$.
(n) $\ln(\cot x)$
(o) $\ln(\sec(2x))$
(p) $f(x) = \ln(\sec x) + \ln(\cot x)$.

40. Differentiate.

(a) 10^{x^3}.
(b) $2^{\tan x}$.
(c) x^x.
(d) x^{x^x}.
(e) $(\sin x)^{\cos x}$.
(f) $(\ln x)^{\ln x}$.

41. Find the limit.

(a) $\lim_{x\to\infty}\left(1 - \frac{2}{x}\right)^x$.
(b) $\lim_{x\to 0}(1-x)^{\frac{1}{x}}$.
(c) $\lim_{x\to\infty}\left(\frac{x}{x-5}\right)^x$.
(d) $\lim_{x\to\infty}\left(\frac{x}{x-2}\right)^{3x+2}$.

42.

(a) Find the linearization of $f(x) = \sqrt{x}$ at $a = 100$ and use it to approximate $\sqrt{99.8}$.
(b) Find the linearization of $f(x) = \sqrt{8+x}$ at $a = 1$ and use it to approximate $\sqrt{9.02}$.
(c) Find the linearization of $f(x) = \sqrt[3]{8+x}$ at $a = 0$ and use it to approximate $\sqrt[3]{7.97}$.

(d) Find the linearization of $f(x) = \ln x$ at $a = 1$ and use it to approximate $\ln 1.01$.

(e) Use a linear approximation to estimate $(1.001)^9$.

(f) Use a linear approximation to estimate $(0.9999)^{2014}$.

43. Express $\frac{\mathrm{d}y}{\mathrm{d}x}$ as a function of x and y by implicit differentiation. The answer key has not been proofread, use with caution.

(a) $x^3 + y^3 = 1$.

(b) $2\sqrt{x} + \sqrt{y} = 3$.

(c) $x^2 + xy - y^2 = 4$.

(d) $2x^3 + x^2y - xy^3 = 2$.

(e) $x^4(x + y) = y^2(3x - y)$.

(f) $y^5 + x^2y^3 = 1 + x^4y$.

(g) $y \cos x = x^2 + y^2$.

(h) $\cos(xy) = 1 + \sin y$.

(i) $4 \cos x \sin y = 1$.

(j) $y \sin\left(x^2\right) = x \sin\left(y^2\right)$.

(k) $\tan\left(\dfrac{x}{y}\right) = x + y$.

(l) $\sqrt{x + y} = 1 + x^2y^2$.

(m) $\sqrt{xy} = 1 + x^2y$.

(n) $x \sin y + y \sin x = 1$.

(o) $y \cos x = 1 + \sin(xy)$.

(p) $\tan(x - y) = \dfrac{y}{1 + x^2}$.

(q) $x^4(x + y) = y^2(3x - y)$.

(r) $2x^3 + x^2y - xy^3 = 2$.

44. Verify that the coordinates of the given point satisfy the given equation. Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

(a) $y \sin(2x) = x \cos(2y)$, $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$.

(b) $\sin(x + y) = 2x - 2y$, (π, π) .

(c) $x^2 + xy + y^2 = 3$, $(1, -2)$ (ellipse).

(d) $x^2 + 2xy - y^2 + x = 2$, $(1, 2)$ (hyperbola).

(e) $y^3 + x^3 + 4xy = \dfrac{3}{4}$, $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

(f) $y^3 + x^3 + 4xy = -4$, $(1, -1)$

(g) $x^2 + y^2 = (2x^2 + 2y^2 - x)^2$, $(0, \frac{1}{2})$.

(h) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$, $(-3\sqrt{3}, 1)$.

(i) $2(x^2 + y^2)^2 = 25(x^2 - y^2)$, $(3, 1)$.

(j) $y^2(y^2 - 4) = x^2(x^2 - 5)$, $(0, -2)$.

(k) $x^{\frac{4}{3}} + y^{\frac{4}{3}} = 10$ at $(-3\sqrt{3}, 1)$.

(l) $x^2y^3 + x^3 - y^2 = 1$ at $(1, 1)$.

45. (a) If V is the volume of a cube with edge length x and the cube expands as time passes, find $\frac{dV}{dt}$ in terms of $\frac{dx}{dt}$.

(b) Each side of a square is increasing at a rate of 1cm/s. At what rate is the area of the square increasing when the area of the square is 9 cm^2?

(c) The radius of a ball is increasing at a constant rate of 5mm/s. At a point in time we measure the radius of the ball to be 5cm.

- How fast is the volume increasing at the time of our observation?
- How fast is the surface area increasing at the time of our observation?

(d) At a point in time, we measure the surface area of a ball to be increasing at a rate of 5cm^2/s and the radius of the ball to be 5cm.

- How fast is the volume increasing at the time of our observation?
- How fast is the radius increasing at the time of our observation?

(e) At a point in time, we measure the volume of a ball to be increasing at a rate of 5cm^3/s and the radius of the ball to be 5cm.

- How fast is the radius increasing at the time of our observation?
- How fast is the surface area increasing at the time of our observation?

(f) A 5m long ladder is leaning on a vertical wall. The base of the ladder is 4m away from the wall. At the moment of observation, the top end of the ladder is sliding downwards along the wall at a speed of 10 cm/s.

- At the time of observation, how fast is the bottom end of the ladder sliding away from the wall
- At the time of observation, how fast is the midpoint of the ladder moving away from the wall?
- At the time of observation, how fast is the midpoint of the ladder moving downwards towards the ground?

(g) A street light is mounted at the top of a 4m tall pole. A woman 160 cm tall walks away from the pole at a speed of 5km/h along a straight path. How fast is the tip of her shadow moving when she is 10m from the pole?

(h) A ship is pulled into a dock. On the dock, the rope is pulled by a pulley that is 1m lower than the mooring point on the ship. If the rope is pulled in at a constant rate of 10cm/s, how fast is the mooring point approaching the dock when it is 10m from the dock?

(i) A Ferris wheel with a radius of 10m is rotating at a rate of one revolution every 2 minutes. How fast is a riding rising when his seat is 16 m above ground level?

(j) The minute hand on a watch is 10mm long and the hour hand is 5mm long. How fast is the distance between the tips of the hands changing at two o'clock?

46. Use the Intermediate Value theorem and the Mean Value Theorem/Rolle's Theorem to prove that the function has **exactly one** real root.

(a) $f(x) = x^3 + 4x + 7$.

(b) $f(x) = x^3 + x^2 + x + 1$.

(c) $f(x) = \cos^3\left(\frac{x}{3}\right) + \sin x - 3x$.

47. Use the Intermediate Value theorem and the Mean Value Theorem/Rolle's Theorem to prove that the function has **exactly one** real root.

(a) $x^5 + 7x = 2$.

(b) $x^7 + x^5 + x^3 = 3$.

(c) $2x - 1 = \sin x$.

(d) $e^x + 2x = 3$.

48. Find the critical points of the function. Identify whether those are local maxima, minima, or neither. The answer key has not been proofread, use with caution.

(a) $f(x) = \dfrac{x}{1+x^2}$.

(b) $f(x) = x^3 - x^2 - x - 1$.

(c) $f(x) = 2x^3 - x^2 - 20x + 1$.

(d) $f(x) = x + \dfrac{1}{x}$.

(e) $f(x) = \dfrac{x - \frac{1}{2}}{x^2 - 2x + \frac{7}{4}}$.

49. Find the maximum and minimum values of f on the given interval and the values of x for which they are attained.

(a) $f(x) = 9 + 3x - x^2$, $x \in [0, 4]$.

(b) $f(x) = 5 + 4x - 2x^3$, $x \in [-1, 1]$.

(c) $f(x) = 2x^3 - x^2 - 20x + 1$, $x \in [-4, 3]$.

(d) $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$, $x \in [-2, 3]$.

(e) $f(x) = x^3 - x^2 - x + 1$, $x \in [-1, 1]$.

(f) $f(x) = x^3 - x + 1$, $x \in [-2, 1]$.

(g) $f(x) = (x^2 - 1)^3$, $x \in [-1, 2]$.

(h) $f(x) = x + \dfrac{1}{x}$, $x \in [0.2, 4]$.

(i) $f(x) = \dfrac{x}{x^2 - x + 1}$, $x \in [0, 3]$.

(j) $f(t) = t\sqrt{4 - t^2}$, $x \in [-1, 2]$.

(k) $f(t) = \sqrt[3]{t}(8 - t)$, $x \in [0, 8]$.

(l) $f(t) = 2\cos t + \sin(2t)$, $x \in [0, \frac{\pi}{2}]$.

(m) $f(t) = t + \cot\left(\dfrac{t}{2}\right)$, $x \in [\frac{\pi}{4}, \frac{7\pi}{4}]$.

(n) $f(t) = t + \cot\left(\dfrac{t}{2}\right)$, $x \in [\frac{\pi}{4}, \frac{7\pi}{4}]$.

(o) $f(x) = xe^{3x}$, $x \in \left[-3, \frac{1}{6}\right]$.

(p) $f(x) = (x-2)(x+1)e^x$, $x \in [-5, 2]$.

(q) $f(x) = (x+1)e^{-x^2}$, $x \in [-3, 3]$.

(r) $f(x) = xe^{2x}$, $x \in \left[-2, \frac{1}{2}\right]$.

50. (a) Find the dimensions of a rectangle with area 1000 m^2 whose perimeter is as small as possible.

(b) A box with an open top is to be constructed from a square piece of cardboard, 1m wide, by cutting out a square from each of the four corners and bending up the sides. Find the largest volume that such a box can have.

(c) A right circular cylinder is inscribed in a sphere of radius r. Find the largest possible volume of such a cylinder.

(d) A wedge of radius 2 (depicted below) is folded into a cone cup. The volume varies depending on the angle of the wedge. Find the maximal possible volume of the cone cup and the angle of the wedge for which this maximal volume is achieved.

51. (a) What is the x-coordinate of the point on the hyperbola $x^2 - 4y^2 = 16$ that is closest to the point $(1, 0)$?

(b) What is the x-coordinate of the point on the ellipse $x^2 + 4y^2 = 16$ closest to the point $(1, 0)$?

(c) A rectangular box with a square base is being built out of sheet metal. 2 pieces of sheet will be used for the bottom of the box, and a single piece of sheet metal for the 4 sides and the top of the box. What is the largest possible volume of the resulting box that can be obtained with $36m^2$ of metal sheet?

(d) Recall that the volume of a cylinder is computed as the product of the area of its base by its height. Recall also that the surface area of the wall of a cylinder is given by multiplying the perimeter of the base by the height of the cylinder.

A cylindrical container with an open top is being built from metal sheet. The total surface area of metal used must equal $10m^2$. Let r denote the radius of the base of the cylinder, and h its height. How should one choose h and r so as to get the maximal possible container volume? What will the resulting container volume be?

52.

Match each of the following function plots:

1. 2. 3. 4.

to their derivative plots:

(a) (b) (c) (d)

Give reasons for your choices. Can you guess formulas that would give a similar (or precisely the same) graph, and confirm visually your guess using a graphing device?

53.

Match each of the following function plots:

1. 2. 3. 4. 5.

to their derivative plots:

(a) (b) (c) (d) (e)

54. Find the

- the implied domain of f,
- x and y intercepts of f,
- horizontal and vertical asymptotes,
- intervals of increase and decrease,
- local and global minima, maxima,
- intervals of concavity,
- points of inflection.

Label all relevant points on the graph. Show all of your computations.

(a) $f(x) = \dfrac{x+\frac{1}{2}}{x^2+x+1}$

(b) $f(x) = \dfrac{2x^2-5x+\frac{9}{2}}{x^2-3x+3}$. **For this problem, indicate only the x-coordinates of the local maxima/minima and inflection points; you do not need to compute the y-coordinates of those points.**

Computation shows that $f'(x) = \dfrac{-x^2+3x-\frac{3}{2}}{(x^2-3x+3)^2}$ and that $f''(x) = \dfrac{(2x-3)x(x-3)}{(x^2-3x+3)^3}$; you may use those computations without further justification.

(c) $f(x) = \dfrac{2\sqrt{-x^2+1}+1}{\sqrt{-x^2+1}+1}$, $f(x) = \dfrac{1}{\sqrt{-x^2+1}+1}$

The two functions are plotted simultaneously in the x, y-plane. Indicate which part of the graph is the graph of which function.

(d) $f(x) = \dfrac{e^x+e^{-x}}{e^x-e^{-x}}$

(e) $f(x) = \dfrac{-e^{-x}+e^x}{e^{-x}+e^x}$

(f) $f(x) = \ln\left(\dfrac{x+1}{-x+1}\right)$

(g) $f(x) = \dfrac{x^2+3x+1}{x^2+2x}$. **For this problem, indicate only the x-coordinates of the local maxima/minima and inflection points; you do not need to compute the y-coordinates of those points.**

Computation shows that $f'(x) = \dfrac{-x^2-2x-2}{(x^2+2x)^2}$ and that $f''(x) = \dfrac{2x^3+6x^2+12x+8}{(x^2+2x)^3} = \dfrac{(x+1)\left(2x^2+4x+8\right)}{(x^2+2x)^3}$; you may use those computations without further justification.

(h) $f(x) = \dfrac{x+1}{x^2+2x+4}$

(i) $f(x) = \dfrac{3x^3 - 30x^2 + 97x - 99}{x^2 - 6x + 8}$. **For this problem, do not find the x-intercepts of the function. Indicate only the x-coordinates of the local maxima/minima and inflection points; you do not need to compute the y-coordinates of those points.**

Computation shows that $f'(x) = \dfrac{3x^4 - 36x^3 + 155x^2 - 282x + 182}{(x^2 - 6x + 8)^2} = \dfrac{(x^2 - 6x + 7)(3x^2 - 18x + 26)}{(x^2 - 6x + 8)^2}$ and that $f''(x) = \dfrac{2x^3 - 18x^2 + 60x - 72}{(x^2 - 6x + 8)^3} = \dfrac{(x-3)(2x^2 - 12x + 24)}{(x^2 - 6x + 8)^3}$; you may use those computations without further justification.

55. (a) Sketch the graph of $y = x^4 - 8x^2 + 8$ by determining the intervals of increase and decrease, finding the local mins and maxes, determining where the graph is concave up and concave down, and plotting a few key points.

(b) Sketch the graph of $y = \frac{x-1}{x^2-9}$ by graphing any vertical and horizontal asymptotes, finding the x- and y-intercepts, and then sketching a graph that fits this information.

(c) Consider the function $f(x) = \dfrac{4x^2 + 10x + 5}{2x + 1}$. Computation shows that $f'(x) = \dfrac{8x^2 + 8x}{(2x+1)^2}$ and $f''(x) = \dfrac{8}{(2x+1)^3}$.

- Find the intervals of increase and intervals of decrease of f.
- Find the local maxima and minima of f.
- Find where the function is concave up and where it is concave down.
- Sketch the function $f(x)$ roughly by hand. Make sure that your plot matches your computations from the preceding parts of the problem.

  You may use the provided grid and coordinate system. From the previous page, we recall that $f(x) = \dfrac{4x^2 + 10x + 5}{2x + 1}$, $f'(x) = \dfrac{8x^2 + 8x}{(2x+1)^2}$ and $f''(x) = \dfrac{8}{(2x+1)^3}$.
  The 4 points plotted on the grid are known to lie on the curve.

(d) Consider the function $f(x) = \dfrac{2x^2 - 4x + 2}{x^2 - 2x}$.

- Find the vertical asymptotes of f. **For this particular sub-question, and for this sub-question alone, no justification is required (just write the answer).**
- Computation shows that $f'(x) = \dfrac{-4x + 4}{(x^2 - 2x)^2}$. Find the intervals of increase and decrease of f.
- Find the local maxima and minima of f.
- Computation shows that $f''(x) = \dfrac{12x^2 - 24x + 16}{(x^2 - 2x)^3}$. Find where the function is concave up and where it is concave down.
- Sketch the function $f(x)$ roughly by hand. Make sure that your plot matches your computations from the preceding parts of the problem.
  You may use the provided grid and coordinate system. We recall that
  $f(x) = \dfrac{2x^2 - 4x + 2}{x^2 - 2x}$,
  $f'(x) = \dfrac{-4x + 4}{(x^2 - 2x)^2}$,
  $f''(x) = \dfrac{12x^2 - 24x + 16}{(x^2 - 2x)^3}$.
  The points plotted below are known to lie on the curve.

56. Estimate the integral using a Riemann sum using the indicated sample points and interval length.

(a) $\displaystyle\int_0^4 \left(\sqrt{8x+1}\right) \mathrm{d}x$. Use four intervals of equal width, choose the sample point to be the left endpoint of each interval.

(b) $\displaystyle\int_0^6 \frac{1}{x^2+1} \mathrm{d}x$. Use three intervals of equal width, choose the sample point to be the left endpoint.

(c) $\displaystyle\int_{-3.5}^{-0.5} \frac{\mathrm{d}x}{x^2+1}$. Use three intervals of equal width, choose the sample point to be the midpoint of each interval.

(d) $\displaystyle\int_0^2 \frac{\mathrm{d}x}{1+x+x^3}$. Use $\Delta x = \frac{1}{2}$ and right endpoint sampling points.

(e) $\displaystyle\int_{-2}^0 \frac{\mathrm{d}x}{1+x+x^2}$. Use $\Delta x = \frac{2}{3}$ and left endpoint sampling points.

(f) $\displaystyle\int_0^2 \frac{\mathrm{d}x}{1+x^3}$. Use four intervals of equal width, choose the sample point to be the left endpoint of each interval.

(g) $\displaystyle\int_{-2}^0 \frac{\mathrm{d}x}{x^4+1}$. Use four intervals of equal width, choose the sample point to be the right endpoint.

(h) $\displaystyle\int_{-1}^0 \frac{1}{3x^2+1}\mathrm{d}x$. Use 3 **intervals** of equal width, choose the sampling points to be the **left endpoints** of each interval. Simplify your answer to a rational number (single fraction of two integers).